

ORIGINAL ARTICLE

Immune dysregulation in early childhood autism: A case control study of pro and anti inflammatory cytokines

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ABSTRACT

Background and aim: Immune dysregulation has been proposed as a contributing mechanism in autism spectrum disorder (ASD). Interleukin-6 (IL-6) and interleukin-10 (IL-10) are key pro- and anti-inflammatory cytokines that may influence early childhood development. This study aimed to compare IL-6 and IL-10 levels between children with ASD and typically developing controls and to assess the association of cytokines level with the severity of autism.

Methods: A case-control study was conducted among children aged 2–5 years with ASD diagnosed according to the DSM-5 criteria and age-matched typically developing controls in Surabaya, Indonesia. Blood serum IL-6 and IL-10 levels were measured via enzyme-linked immunosorbent assay (ELISA). ASD severity was assessed via the Childhood Autism Rating Scale, second edition (CARS-2). Nonparametric statistical analyses were applied according to the data distribution.

Results: A total of 70 participants were included. Children with ASD (n = 35) presented significantly higher IL-6 levels (1.21 vs 0.60 pg/mL, $P < 0.001$) than the control group even after adjustment for sex and age. IL-10 levels did not differ significantly between the groups ($P = 0.165$). There was a moderate association between IL-6/IL-10 ratio with ASD severity in unadjusted analyses; however, this association did not remain statistically significant after adjustment for age and sex.



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Conclusions: Children with ASD demonstrated elevated IL-6 levels compared with typically developing controls. Evidence supporting broader immune imbalance or independent associations with symptom severity remains preliminary. Further studies are needed to clarify the role of inflammatory signaling in early childhood ASD. (www.actabiomedica.it)

Key words: autism, IL-6, IL-10, neuroinflammation, early childhood development, mental health

Introduction

Autism spectrum disorder (ASD) is a heterogeneous neurodevelopmental condition characterized by persistent impairments in social communication and restricted, repetitive behavior, with onset in early childhood (1,2). Despite advances in diagnostic recognition and support services, ASD remains associated with substantial lifelong functional limitations and significant psychosocial and economic burden for affected individuals and their families (1,2). The marked clinical heterogeneity of ASD suggests that multiple biological pathways contribute to its pathogenesis, yet the mechanisms underlying this variability remains incompletely understood (2,3). Accumulating evidence indicates that immune dysregulation represents a key biological pathway through which genetic susceptibility and environmental exposure may converge to influence early brain development (3). Interest in immune involvement in ASD dates back nearly five decades, when Stubbs first reported altered immune responsiveness in children with autism, including the absence of rubella antibody titers despite routine vaccination and reduced mitogen-induced peripheral blood mononuclear cell activity. Since then, a growing body of experimental, clinical, and epidemiological research has supported the role of both prenatal and postnatal immune disturbances in ASD, suggesting that altered immune signaling is not merely a secondary phenomenon but may contribute directly to neurodevelopmental vulnerability (4,5).

Cytokines play a central role in the intersection of immune function and neurodevelopment. In addition to their classical role as immune signaling molecules, cytokines are actively involved in neuronal

differentiation, synaptic pruning, and circuit maturation within the central nervous system (6,7). Among these, interleukin-6 (IL-6) and interleukin-10 (IL-10) are of particular interest because of their opposing roles in inflammatory regulation. IL-6 is a pleiotropic proinflammatory cytokine produced primarily by innate immune cells, including microglia, and has been shown to influence synaptic organization, excitatory-inhibitory balance, and blood-brain barrier integrity (8–11). Elevated IL-6 levels have been reported in postmortem brain tissue, cerebrospinal fluid, and peripheral blood of individuals with ASD, and experimental models have demonstrated that excessive IL-6 signaling during early development can disrupt neurodevelopmental trajectories relevant to ASD phenotypes (3,12). However, findings across clinical studies remain inconsistent, as several reports have found no significant difference in IL-6 levels between children with ASD and typically developing controls (13,14). Conversely, IL-10 serves as an anti-inflammatory and immunoregulatory cytokine that constrains excessive immune activation and promotes immune homeostasis. Reduced IL-10 levels and impaired anti-inflammatory signaling have been reported in subsets of individuals with ASD, suggesting a diminished capacity to resolve inflammation (15). In contrast, other studies have reported higher IL-10 levels in children or adolescents with ASD, further highlighting heterogeneity in immune findings across studies (16). These divergent findings across studies underscore the complexity and heterogeneity of immune alterations in ASD. Emerging evidence indicates that the balance between pro- and anti-inflammatory signaling—rather than absolute levels of individual cytokines—may be more biologically and clinically relevant. Ratios such as the

IL-6/IL-10 ratio have been proposed as integrative markers of immune imbalance and are associated with behavioral severity and neurodevelopmental outcomes in individuals with ASD (12,17).

Despite substantial progress in delineating immune alterations in individuals with ASD, most studies have been conducted in high-income countries, limiting the generalizability of findings across diverse populations. Children in low- and middle-income countries may experience distinct environmental exposures, nutritional profiles, infectious burdens, and healthcare access, all of which can influence immune development and inflammatory regulation (18). However, data from these settings remain markedly underrepresented in ASD research, particularly during early childhood, a critical period of concurrent brain and immune system maturation (19). Taken together, these inconsistencies suggest that cytokine alterations in ASD may vary across populations, age groups, clinical phenotypes, and methodological approaches. Therefore, investigating cytokine profiles in different geographic and ethnic populations, including underrepresented regions such as Indonesia, may help clarify the variability of immune findings in ASD. Accordingly, the present study aimed to characterize inflammatory profiles in young children with ASD by examining serum concentrations of IL-6 and IL-10, as well as their ratio, in comparison with typically developing controls in an Indonesian population. We hypothesized that an imbalance between pro- and anti-inflammatory signaling, reflected by the IL-6/IL-10 ratio, would be associated with ASD severity. We also explored the relationships between these immune markers and autism symptom severity. By providing region-specific data from a middle-income country and focusing on early childhood, this study aims to contribute to a more globally representative understanding of immune dysregulation in autism.

Patients and Methods

Study design and participants

This case-control study was conducted at the pediatric and psychiatric clinics of Dr. Soetomo

General Academic Hospital and Pusura Tegalsari Women and Children Hospital, Surabaya, Indonesia, between September and December 2025. This study included 70 children aged 24–59 months, comprising 35 children with autism spectrum disorder (ASD) and 35 typically developing (TD) controls.

Children in the ASD group were eligible if they (1) had a diagnosis of ASD established according to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), confirmed by a board-certified neuropsychiatrist or psychiatrist, and (2) were between 24 and 59 months of age. Autism symptom severity was assessed using the Childhood Autism Rating Scale, second edition (CARS-2), which yields a total score ranging from 15–60, with higher scores indicating greater symptom severity (20,21).

Control participants were included if they (1) demonstrated typical development based on the Denver Developmental Screening Test II (Denver II) and the Strength and Difficulties Questionnaire (SDQ) and (2) were aged 24–59 months. Typically developing children were screened to exclude those with a history of neurological, psychiatric, or developmental disorders and were unmedicated and in good health at the time of participation. The exclusion criteria for both groups included the following: (1) clinically apparent syndromic features; (2) a history of autoimmune disease, acute infection, or allergic conditions; (3) receipt of vaccination within a defined period prior to enrolment; and (4) severe malnutrition (22).

Ethical approval and consent to participate

The study protocol was approved by the Ethics Committee of Dr. Soetomo General Academic Hospital (approval number: 1403/KEPK/VIII/2025). Written informed consent was obtained from the parents or legal guardians of all participants prior to enrolment.

Sample size rationale

Given the limited availability of prior data on IL-6 and IL-10 profiles in young children with autism spectrum disorder from low- and middle-income countries, this study was designed as an exploratory case-control investigation. The sample size was determined based

on feasibility within the study period while remaining comparable to previous cytokine studies in early childhood ASD populations. Sample size was estimated for the comparison of two independent means using a two-sided α of 0.05 and a statistical power of 80% ($\beta = 0.20$), based on the expected mean difference and standard deviations reported by Han et al. With an allocation ratio of 1:1, the minimum required sample size was 30 participants per group. This sample size was considered adequate to detect moderate between-group differences in inflammatory markers and to explore associations with symptom severity (23).

Biospecimen collection and cytokine assay

Peripheral venous blood samples (6 mL) were collected between 08:00 and 12:00 in tubes with a clotting activator. The samples were centrifuged at 3000 rpm for 10 minutes within 30 minutes of collection, transported under temperature-controlled conditions, and stored at -20°C for no longer than 6 months prior to analysis.

Serum concentrations of IL-6 and IL-10 were measured via commercially available enzyme-linked immunosorbent assay (ELISA) kits (Human IL-6 ELISA Kit, Elabscience, Wuhan, China; Catalog No: E-EL-H6156; and Human IL-10 ELISA Kit, Elabscience, Wuhan, China; Catalog No: E-EL-H6154), with a reported sensitivity of 0.94 pg/mL for both assays. All procedures were performed according to the manufacturer's instructions at the Institute of Tropical Disease Laboratory, Surabaya, Indonesia. Cytokine concentrations were calculated via standard curves. Samples with cytokine concentrations below the assay limit of detection (LOD) were substituted as LOD/2 (0.47 pg/mL), allowing inclusion of all participants in analyses involving IL-10 and the IL-6/IL-10 ratio (24).

Statistical analysis

Data distribution was assessed using normality testing. Continuous variables were summarized as median and interquartile range (IQR) due to non-normal distributions. Group comparisons of demographic variables were performed via independent *t*-test, chi-square test, or Fisher's exact test, as appropriate. Between-group differences in cytokine levels were

analyzed via the Mann-Whitney U test. Quantile regression ($\tau = 0.5$) was conducted to examine differences in inflammatory markers between groups after adjusting for age and sex. Correlations between cytokine levels, IL-6/IL-10 ratio, and autism symptom severity (CARS-2 scores) were evaluated via Spearman's rank correlation coefficient. Adjusted associations with CARS-2 scores were further examined using quantile regression. Given the modest sample size and to avoid model overfitting, covariate adjustment was restricted to a priori confounders (age and sex), which are consistently associated with both cytokine levels and ASD. For quantile regression analyses, standard errors and confidence intervals were estimated via bootstrap resampling with 1,000 replications. All of the statistical analyses were performed using RStudio version 2025.05.1+513. A two-sided *P* value < 0.05 was considered statistically significant.

Results

Demographic comparisons

A total of 70 children were included in the analysis, comprising 35 children with ASD and 35 TD controls. The demographic and clinical characteristics of the study participants are summarized in Table 1. The mean age of children in the ASD group was 47.1 ± 7.9 months, compared with 42.9 ± 9.3 months in the TD group ($P = 0.050$). The ASD group included a higher proportion of males (71.4%) than the TD group (48.6%), although this difference did not reach statistical significance ($P = 0.088$). Differences in pre-term birth (14.3% vs 0%, $P = 0.054$) and pregnancy complications (45.7% vs 22.9%, $P = 0.078$) were also observed but did not achieve conventional levels of statistical significance. No significant differences were observed between the groups with respect to history of low birth weight or perinatal complication. Given the modest sample size, residual imbalance in these characteristics cannot be excluded.

The data are presented as mean $\pm$ standard deviation (SD), median (interquartile range, IQR) or number (percentage), as appropriate. Group comparisons were performed via independent *t*-tests, chi-square tests, or

Table 1. Demographic and clinical characteristics of study participants.

	Case (n=35)	Control (n=35)	P value
Age, months, mean $\pm$ SD	47.1 $\pm$ 7.9	42.9 $\pm$ 9.3	0.050
Sex, n (%)			0.088
Male	25 (71.4%)	17 (48.6%)	
Female	10 (28.6%)	18 (51.4%)	
Low birth weight, n (%)	3 (8.6%)	2 (5.7%)	1
Preterm birth, n (%)	5 (14.3%)	0 (0%)	0.054
Pregnancy complication, n (%)	16 (45.7%)	8 (22.9%)	0.078
Perinatal complication, n (%)	9 (25.7%)	12 (34.3%)	0.602
CARS-2 score, median (IQR)	28.50 (22.00–34.75)	-	-

Table 2. Comparison of inflammatory biomarkers between children with autism spectrum disorder and controls.

Variable	Case (n=35)	Control (n=35)	Mann-Whitney U	Adjusted quantile regression ($\tau = 0.5$)	
			P value	β	P value
IL-6, pg/mL, median (IQR)	1.21 (0.84–1.60)	0.60 (0.49–0.84)	<0.001	0.56	<0.001
IL-10, pg/mL, median (IQR)	0.87 (0.47–3.00)	0.47 (0.47–1.97)	0.165	0.40	0.251
IL-6/IL-10 Ratio, median (IQR)	1.16 (0.57–1.69)	1.03 (0.34–1.16)	0.099	0.15	0.515

Fisher's exact tests, as appropriate. CARS-2: Childhood Autism Rating Scale, Second Edition. Pregnancy complications include hypertension, preeclampsia, hyperemesis gravidarum, COVID-19, and other causes of hospitalization during pregnancy. Perinatal complications were assessed based on history of perinatal asphyxia and hospitalization in the neonatal intensive care unit.

Differences in serum cytokine levels between groups

The serum concentrations of inflammatory markers comparisons are summarized in Table 2 and presented in Figure 1. Compared with TD controls, children with ASD presented significantly higher serum IL-6 levels (median: 1.21 pg/mL (IQR 0.84–1.60) vs 0.60 pg/mL (IQR 0.49–0.84); $P < 0.001$; Figure 1A). In contrast, the IL-10 concentration did not differ significantly between the ASD and TD groups (median: 0.87 pg/mL (IQR 0.47–3.00) vs

0.47 pg/mL (IQR 0.47–1.97); $P = 0.165$; Figure 1B). To assess the balance between proinflammatory and anti-inflammatory signaling, the IL-6/IL-10 ratio was calculated. The median IL-6/IL-10 ratio was higher in children with ASD than in TD controls, but not significant (median: 1.16, IQR 0.57–1.69 vs 1.03, IQR 0.34–1.16; $P = 0.099$; Figure 1C).

Values are presented as median (interquartile range, IQR). Group comparisons were performed using the Mann–Whitney U test. Quantile regression coefficients (β) represent adjusted differences in the median biomarker levels between groups ($\tau = 0.5$), adjusted for age and sex. Positive β values indicate higher median levels in the ASD group compared with controls.

Adjusted group differences in cytokine concentrations

Quantile regression analysis was performed to further examine the differences in cytokine levels

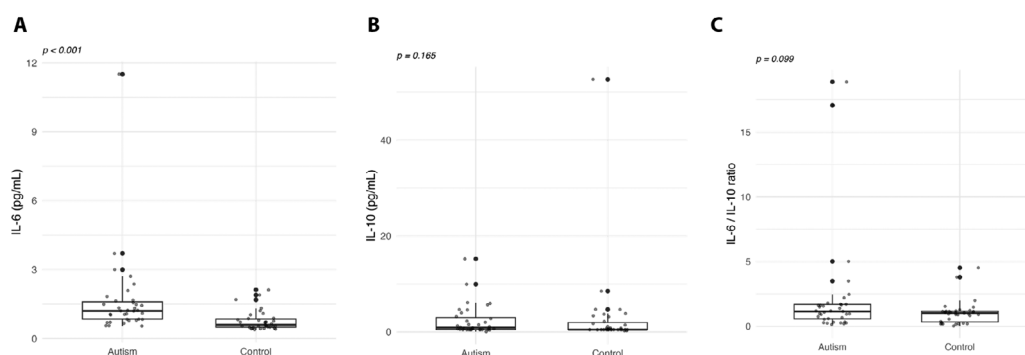


Figure 1. Comparison of the serum concentrations of A) IL-6 and B) IL-10 and the C) IL-6/IL-10 ratio between children with autism spectrum disorder (ASD) and typically developing (TD) controls. Data are presented as medians with interquartile ranges. Group comparisons were performed using the Mann–Whitney U test.

Table 3. Association between inflammatory markers and autism symptom severity (CARS-2 score).

Variable	Spearman correlation		Adjusted quantile regression ($\tau = 0.5$)	
	r	P value	B	P value
CARS-2 vs IL6	0.21	0.227	0.85	0.724
CARS-2 vs IL10	-0.17	0.337	-0.69	0.431
CARS-2 vs IL6/IL10 Ratio	0.36	0.032	0.64	0.621

between groups after adjustment for age and sex. The results were consistent with the unadjusted analysis, with the levels of IL-6 ($\beta = 0.56$, 95% CI = [0.27; 0.86], $P < 0.001$) remaining significantly greater in the autism group than in the control group, whereas the level of IL-10 ($\beta = 0.399$, 95% CI = [-0.28; 1.08], $P = 0.251$) and the IL-6/IL-10 ratio ($\beta = 0.145$, 95% CI = [-0.29; 0.58], $P = 0.515$) were not significantly different between the groups.

Adjusted correlation between inflammatory markers and autism symptom severity

The median CARS-2 score among children with ASD was 28.50 (IQR 22.00–34.75). Quantile regression analyses were performed to examine the associations between inflammatory markers and autism symptom severity after adjustment for age and sex (Table 3). In adjusted models, neither IL-6 ($\beta = 0.85$, 95% CI = [-3.80; 5.51], $P = 0.72$) nor IL-10 ($\beta = -0.69$, 95% CI = [-2.38; 1.00], $P = 0.43$) was significantly associated with CARS-2 scores. Similarly, the IL-6/IL-10 ratio was not significantly associated

with autism severity ($\beta = 0.64$, 95% CI = [-1.89; 3.18], $P = 0.62$). The wide confidence intervals indicate limited statistical precision, suggesting that these findings should be interpreted cautiously.

Spearman correlation coefficients represent unadjusted monotonic associations between biomarkers and Childhood Autism Rating Scale (CARS)-2 scores. Adjusted quantile regression coefficients (β) represent the association with the median CARS-2 score ($\tau = 0.5$), adjusted for age and sex. CARS-2: Childhood Autism Rating Scale, Second Edition.

In the unadjusted analysis, Spearman correlation analyses between inflammatory markers and autism symptom severity are illustrated in Figure 2. The serum IL-6 level demonstrated a weak positive correlation with CARS-2 score ($r = 0.21$, $P = 0.227$), which did not reach statistical significance (Figure 2A). Serum IL-10 showed a weak negative correlation with CARS-2 score ($r = -0.17$, $P = 0.337$), which was also not statistically significant (Figure 2B). The IL-6/IL-10 ratio was moderately positively correlated with CARS-2 score in unadjusted analysis ($r = 0.36$, $P = 0.032$; Figure 2C).

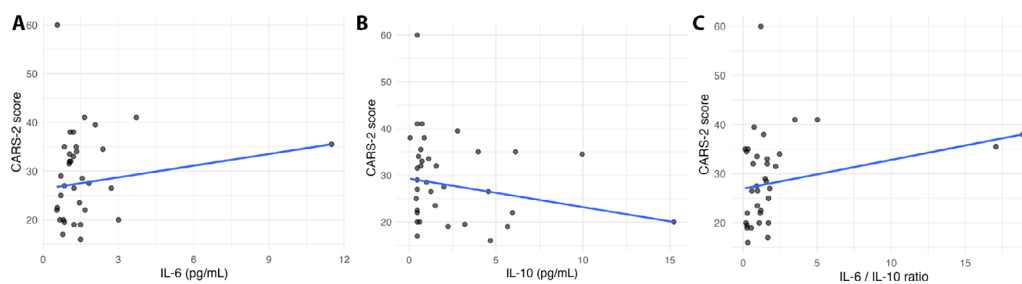


Figure 2. Scatter plots illustrating the associations between CARS-2 scores and A) serum IL-6 levels, B) serum IL-10 levels, and C) the IL-6/IL-10 ratio in children with autism spectrum disorder. Scatter plots show individual data points. The solid lines indicate unadjusted median trends estimated via quantile regression.

Discussion

Principal findings

In this case-control study of young children with autism spectrum disorder from Indonesia, we observed significantly elevated serum IL-6 levels compared with those of typically developing controls, while the IL-10 levels did not differ significantly between groups. The IL-6/IL-10 ratio demonstrated an association with autism symptom severity in unadjusted analyses, however, this association was attenuated after adjustment for age and sex. These findings suggest that elevated pro-inflammatory signaling, particularly IL-6, is associated with early childhood ASD.

Proinflammatory signaling and IL-6 in ASD

The observed increase in IL-6 in children with ASD is aligned with a body of literature suggesting involvement of proinflammatory signaling in subsets of individuals with autism (16,25). Several studies and meta-analyses have reported elevated serum or plasma IL-6 levels in ASD, however, findings have not been uniform, with some cohorts demonstrating no significant difference or even reduced levels (12–14,23,26,27). In addition to peripheral findings, experimental and postmortem studies have demonstrated that IL-6 is elevated within the autistic brain and can directly influence neurodevelopment by disrupting neuronal migration, synaptic organization, and the excitatory-inhibitory balance (8).

IL-6 is produced primarily by innate immune cells, including microglia, and plays a dual role in

immune activation and neural signaling (6). Under inflammatory conditions, IL-6 has been shown to alter blood-brain barrier permeability and facilitate immune-brain communication, providing a plausible mechanistic link between peripheral inflammation and central neuroimmune activation (28,29). Sex-specific patterns of IL-6 dysregulation have been described in some studies, with higher levels reported in boys and variable findings in girls (30). In our study, although sex distribution differed between groups, adjustment for sex did not materially alter the association between ASD status and IL-6 levels, suggesting that the observed elevation was not solely attributable to sex differences. Nevertheless, the heterogeneity of IL-6 findings across studies, including reports of no difference or reduced IL-6 levels in specific populations, highlights the context-dependent nature of cytokine regulation and underscores the need to interpret IL-6 alterations within broader immune networks rather than as standalone markers (7,31,32).

Anti-inflammatory regulation and IL-10

IL-10 serves as a key counterregulatory cytokine that limits excessive inflammatory responses and maintains immune homeostasis (15). In this study, a proportion of IL-10 samples yielded concentrations below the assay LOD, reflecting the very low circulating levels typically observed in young children in the absence of acute inflammation and infection. As a result, reduced analytical sensitivity and measurement variability may have limited the ability to detect subtle between-group differences. IL-10 levels did not differ

significantly between the ASD and control groups, and no independent association with symptom severity was observed after adjustment. Prior studies have reported inconsistent IL-10 findings in individuals with ASD, with both reduced and elevated levels described, suggesting that variability is related to age, sex, disease stage, and immune context (7,16,33). Given the low baseline concentration of IL-10 and the proportion of values below the assay LOD, these findings should be interpreted cautiously. IL-10 is a tightly regulated anti-inflammatory cytokine and is typically present at very low circulating concentrations in healthy children (34).

Rather than reflecting a simple deficiency of anti-inflammatory signaling, unchanged or variably altered IL-10 levels may indicate an insufficient compensatory response to sustained proinflammatory activation (35). This interpretation aligns with models of chronic immune imbalance in which regulatory mechanisms are present but inadequate to restore homeostasis, particularly during critical periods of neurodevelopment (36). Although altered anti-inflammatory regulation has been proposed in ASD, our data do not provide definitive evidence of impaired IL-10-mediated compensation in early childhood. Further studies using more sensitive assay platforms and longitudinal designs may help clarify the role of IL-10 in neurodevelopmental immune regulation.

The IL-6/IL-10 ratio and immune imbalance

Although the IL-6/IL-10 ratio was numerically higher in the ASD group, the between-group difference did not reach statistical significance. In unadjusted analysis, the IL-6/IL-10 ratio showed an association with autism symptom severity; however, this association was attenuated after adjustment for age and sex. These findings suggest that examining the balance between pro- and anti-inflammatory pathways may provide additional biological insight beyond absolute cytokine concentrations, although this relationship requires further confirmation in larger cohorts. Prior studies have reported associations between inflammatory profiles and behavioral impairment, sleep disturbance, and social dysfunction in individuals with ASD (37). The attenuation of this association after

covariate adjustment suggests that the relationship between immune imbalance and symptom severity may be influenced by developmental or demographic factors during early childhood.

From a neuroinflammatory perspective, a higher IL-6/IL-10 ratio has been proposed as a potential indicator of relative proinflammatory predominance. This imbalance may reflect sustained microglial activation or altered innate immune signaling that disrupts synaptic refinement and neural circuit maturation (29). This microglia-mediated disruption of synaptic organization and excitatory–inhibitory balance has been proposed as a plausible biological link between immune dysregulation and autism-related behavioral phenotypes (3,8,38). Given that cytokines and their receptors are expressed by neurons and glial cells, prolonged skewing toward proinflammatory signaling during early childhood could have lasting effects on brain development and behavior (39). However, given the absence of statistical significance in our adjusted analysis, this interpretation should be considered exploratory.

Peripheral biomarkers and central neuroinflammation

A major limitation of this study is the reliance on peripheral blood cytokine measurements, which may not directly reflect inflammatory processes within the central nervous system. Previous studies have demonstrated both correspondence and dissociation between peripheral and cerebrospinal fluid cytokine profiles in ASD, highlighting the complexity of immune–brain interactions (40). Phenotypic shifts in microglia triggered by peripheral inflammation may therefore profoundly affect neuronal signaling and brain information transmission (41). Nevertheless, experimental evidence indicates that peripheral immune activation can influence central neuroinflammatory responses through blood–brain barrier modulation, endothelial signaling, and microglial priming (29).

Recent reports of elevated IL-6 levels in the cerebrospinal fluid of children with ASD provide further support for the relevance of central immune activation, even when peripheral markers are used as indirect indicators (40). Accordingly, peripheral cytokines should

be interpreted as proxies of systemic immune states that may interact with central neuroimmune mechanisms rather than as definitive measures of brain inflammation (38).

Strengths, limitations, and future directions

This study has several strengths, including its focus on a narrowly defined early childhood age range corresponding to a critical window of neuroimmune development, the simultaneous assessment of pro- and anti-inflammatory cytokines, and the inclusion of participants from a middle-income country, addressing the underrepresentation of such populations in ASD immunological research.

Limitations include the cross-sectional design, which precludes causal inference, the modest sample size, detection-limit handling bias, group-mismatch confounding, limited covariate control, and the absence of direct central nervous system immune measurements. Future studies employing longitudinal designs, larger samples, integrating peripheral biomarkers with neuroimaging or genetic data, and ASD subgroups on the basis of immune profiles will be essential to clarify the temporal and mechanistic relationships between immune imbalance and neurodevelopmental outcomes (42).

Conclusions

In conclusion, our findings indicate that children with ASD exhibit higher serum IL-6 levels than typically developing controls, suggesting a possible pro-inflammatory shift during early childhood. Although IL-6/IL-10 ratio showed an association with autism symptom severity in unadjusted analyses, this relationship did not remain statistically significant after adjustment for age and sex, indicating that evidence for broader immune imbalance remains preliminary. Taken together, these findings support the relevance of inflammatory signaling in ASD while underscoring the need for cautious interpretation. Replication in larger, longitudinal, and more comprehensively controlled cohorts will be necessary to clarify the stability and clinical significance of cytokine alterations in early childhood autism.

Ethical Approval: The study protocol was approved by the Ethics Committee of Dr. Soetomo General Academic Hospital (approval number: 1403/KEPK/VIII/2025)

Conflict of Interest: The authors declare that they have no competing interest.

Authors' Contribution: S.M.S. conceived the study, contributed to data collection and curation, and drafted the manuscript. P.I.G. and Y.S. contributed to study design and supervision. N.D.A. performed the statistical analysis and visualization. R.N. contributed to data collection and curation. A.D.W.W. provided immunology supervision. All authors reviewed and approved the final manuscript.

Declaration on the Use of AI: None.

Consent for publication: Not applicable

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